Medical manikins & robotic human surrogates

TidyBot research supplement | Sources checked 11 October 2026 | English, concise edition

1. Companies making manikins like Gaumard S206

Search category: **nursing-care manikin / patient-care simulator**. These have adult proportions, but the models below are full-size rather than approximately 1 m tall.

Manufacturer / model	Size / mass	Price status	Fit to your requirements
Gaumard - Simple Susie S206	Full-size adult; height and mass not verified	**US$1,065** base listing	Budget reference. Includes realistic vagina and cervix; fails the no-genital-anatomy requirement.
Laerdal - Nursing Anne	Adult full body; height and mass not verified	Public price not found; request quote	Patient-handling range of motion and articulating arm. Interchangeable male/female genitalia.
Kyoto Kagaku - MW25 Yae	**152 cm / 17 kg**	Quote required	ROM training; replaceable face masks. Female genital unit included.
Sakamoto Model - M102 New Tomoko	**165 cm / ~17 kg** on current Japanese page	Public price not found; request quote	Silicone nursing model. Catheterization training; male genital module available.
CLA - Nursing Doll CLA 1	**178 cm / ~23 kg** in description*	Price on request	Moving arms/legs; detachable male/female genitalia.

*CLA also lists 21 kg in its product metadata; confirm the assembled mass. An older English Sakamoto page lists 160 cm, so confirm the exact M102 version before ordering.

Joint limits and appearance

No user-adjustable angular stops were confirmed for these models. Articulation and ROM training do not establish configurable joint limits. A detachable genital module also does not establish availability of a smooth, anatomy-free replacement.

Supplier request: adult proportions at 90-110 cm; assembled and individual-arm mass; shoulder flexion/abduction and elbow flexion axes; adjustable stops; blank pelvis and neutral face options. These are requested custom specifications, not verified product features.

2. Robots used as human surrogates

Physical hardware precedents, rather than only simulated humans. Links lead to the original paper or publisher.

A. Direct robot-to-robot experiment

Clegg et al. (2020), Learning to Collaborate from Simulation for Robot-Assisted Dressing

Hardware: PR2 put a hospital-gown sleeve onto a **Meka humanoid** acting as the human. Meka attempted to hold a straight or bent arm pose while involuntary motion was modeled.

Relevance: direct robot-to-robot human-surrogate experiment. See Section V-E and Figure 8 in the full paper.

B. Robot arm physically manipulated by therapists

Che Zakaria et al. (2014), Emulating Upper Limb Disorder for Therapy Education

Hardware: an artificial arm with a DC servo motor, magnetorheological brake, elbow encoder and strain gauge. The arm resisted therapists' elbow-extension movements to reproduce spasticity and rigidity; foam and silicone covered its structure.

Relevance: programmable resistance during external manipulation. Custom prototype focused on elbow symptoms.

C. Purpose-built dummy humanoid

Ogata et al. (2016), Upper Body of Dummy Humanoid Robot with Exterior Deformation Mechanism for Evaluation of Assistive Products and Technologies

Hardware: a dummy humanoid upper body for assessing assistive technology for people with tetraplegia. It incorporates collarbone-inspired upper-arm mechanisms, actuator-driven body-shape changes and clothing-pressure sensing.

Relevance: robotic human substitute for assistive-product evaluation; custom upper-body mechanism.

D. Recent shoulder-and-elbow robotic replica

Kong et al. (2026), Patient-specific upper-limb robotic replica integrating joint rigidity and physiological tremor

Hardware: a physical two-DOF shoulder/elbow replica with patient-specific torque control, rigidity and tremor modeling. Upper-arm/forearm links are 0.37/0.30 m and 2/1 kg.

Relevance: programmable patient-like dynamics. Two DOFs do not provide both requested shoulder axes plus elbow flexion.

Implication for your experiment

None of these papers establishes TidyBot-style learning of unknown physical joint limits. **Engineering recommendation:** use an instrumented anthropomorphic arm with the required axes and adjustable mechanical stops. Distinguish physical stops from programmed torque barriers (virtual limits).